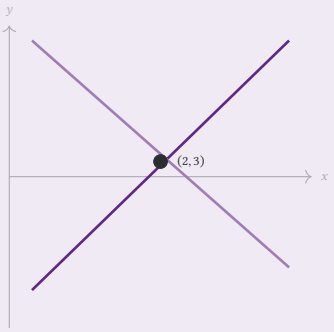
QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 2

RULE · Solution of a system

Solution = the point where both lines cross.

$(2, 3)$: $2 + 3 = 5$ and $2(2) - 3 = 1$ – both true.



RULE · Substitution method

Isolate one variable, substitute into the other equation.

$$y = 2x - 1,\ 3x + y = 9 \implies (x, y) = (2, 3)$$

RULE · Elimination method

Multiply so coefficients match, then add or subtract.

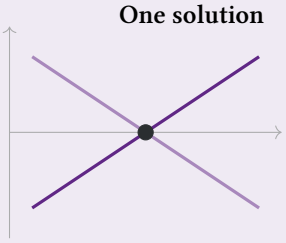
$$x + 2y = 11,\ 3x + y = 18 \implies (x, y) = (5, 3)$$

RULES SHEET – 2/2

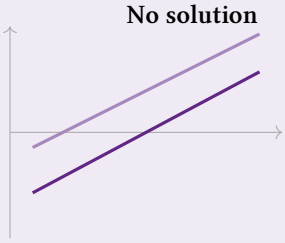
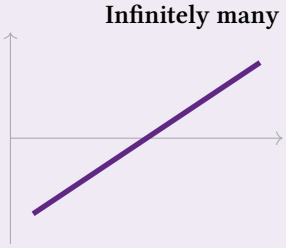
RULE · Special cases

Coefficient ratios	Result
Unequal	One solution
Equal, constants differ	No solution
Equal, constants equal	Infinitely many

One solution



No solution



Infinitely many